

HCSM-48

Faddeev-Popov G_N Normalization

The B_1 Isotypic Selection and the First Principles Origin of $(d+1)^2 = 25$

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Abstract

We derive the normalization factor $(d+1)^2 = 25$ in the HCSM Newton constant from a complete Faddeev-Popov path integral on the discrete torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$. The hidden B_1 isotypic sector of the 56-dimensional Hodge complex at the self-paired eigenvalue $\lambda = 4$ has dimension exactly 5, decomposing as

$$B_1^{\text{hidden}} = \underbrace{3}_{\Omega^1|_{\lambda=4}} \oplus \underbrace{2}_{\Omega^2|_{\lambda=4}} = (d-1) + \frac{d}{2} = d+1.$$

We prove a structural selection theorem: among the five D_4 isotypic sectors of the Hodge complex at $\lambda = 4$, only B_1 satisfies the Faddeev-Popov cancellation identity

$$\det(\Delta - 4I)|_{B_1 \subset \Omega^0, \text{ nonzero}} = \sqrt{\det(D^2|_{B_1^{\text{hidden}}})} = 2^5.$$

The fluctuation determinant on the 8 non- $\lambda = 4$ modes of B_1 in Ω^0 equals exactly the square root of the one-loop determinant on the 5-dimensional hidden B_1 sector. The Faddeev-Popov measure therefore enforces the kinetic term coefficient

$$C_{\text{kin}} = \underbrace{5^2}_{(d+1)^2} \times \underbrace{\frac{2^5}{2^5}}_{\text{FP cancellation}} = 25,$$

giving $G_N = 25/(8\pi M_P^2)$ as a first principles result rather than a fitted normalization. The uniqueness of B_1 is a genuine consequence of the D_4 representation theory of the 8×8 torus: it is the only irrep satisfying $\Omega^0 \text{ mult} = 2 \times \text{hidden mult}$, which is the exact condition required for mirror-pair cancellation. The B_2 sector, despite having the same hidden dimension 5, does not satisfy the identity ($\det \text{ ratio} = 1/2$) and is a spectator in the two-loop renormalization group: $\mathfrak{su}(2)_L$ acts only on B_1 . We separate the derivation into substrate-level theorems, a single disclosed physical-normalization input, and theorems that follow given that input. All numerical claims are verified in reproducible Python at machine precision (IEEE-754 float64).

Derivation status summary.

- DERIVED (substrate-level, no particle identifications, no $N/4$ anchors used): hidden B_1 dimension $3 + 2 = 5$; bilinear trace $5^2 = 25 = (d+1)^2$; FP cancellation identity $\det(\Delta - 4I)|_{B_1 \subset \Omega^0, \text{ nonzero}} = 2^5$; uniqueness of B_1 among the five D_4 irreps; B_2 spectator status; kinetic coefficient $C_{\text{kin}} = 25$.

- DISCLOSED INPUT (physical normalization, flagged in source as carrying a retracted-layer residual): $M_P^2 = N k_L^2 \lambda_L^2$. This single input carries the SYK Lyapunov exponent λ_L , which is a residual of the retired SYK/QIN layer. The framework’s 25 and 8π factors do not depend on this input; only the numerical match to the observed Newton constant does.
- THEOREMS GIVEN THE DISCLOSED INPUT: $G_N = 25/(8\pi M_P^2)$ from first principles, matching observation to 0.14%.

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1 Introduction

1.1 The open problem

The Hodge Complex Standard Model (HCSM) derives Standard Model structure from the spacetime dimension $d = 4$ through the discrete substrate $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ [2, 3].

The graviton sector of the framework was established in HCSM-45 [10]: the graviton is the self-paired curvature mode of the substrate, the $54B_1 \oplus 54B_2$ content of $\text{Sym}^2(\Omega^1|_{\lambda=4})$, with polarization space $V = B_1 \oplus B_2$. HCSM-45 derived the Newton constant form

$$G_N = \frac{(d+1)^2}{8\pi M_P^2}, \quad M_P^2 = N k_L^2 \lambda_L^2, \quad (1)$$

matching observation to 0.13–0.14%. In that paper the factor $(d+1)^2 = 25$ was flagged as *conjectural*: it matched observation but was not derived from first principles.

The present paper completes the path-integral derivation. We establish that the hidden B_1 isotypic sector of the 56-dimensional Hodge complex at the self-paired eigenvalue $\lambda = 4$ is the structural origin of the normalization factor $(d+1)^2 = 25$, and that the Faddeev–Popov measure on the discrete torus enforces this coefficient exactly.

1.2 Summary of results

The present paper establishes the following.

1. (Theorem 3.2). The hidden B_1 isotypic sector of the 56-dimensional Hodge complex at $\lambda = 4$ is exactly 5-dimensional, decomposing as $3\Omega^1 \oplus 2\Omega^2 = (d-1) \oplus (d/2) = d+1$, with bilinear trace $5^2 = 25$.
2. (Theorem 4.1). The Faddeev–Popov fluctuation determinant on the B_1 isotypic sector of Ω^0 satisfies the exact identity $\det(\Delta - 4I)|_{B_1 \subset \Omega^0, \text{nonzero}} = 2^5 = 32$.
3. (Theorem 5.1). Among the five D_4 irreps of the Hodge complex at $\lambda = 4$, only B_1 satisfies the Faddeev–Popov cancellation identity

$$\det(\Delta - 4I)|_{\rho \subset \Omega^0} = \sqrt{\det(D^2|_{\rho^{\text{hidden}}})}.$$

For A_1, A_2, B_2, E , the ratios are 64, $-1/8$, $1/2$, and 128 respectively.

4. (Theorem 6.1). The kinetic-term coefficient is $C_{\text{kin}} = 5^2 \cdot (2^5/2^5) = 25$, and therefore $G_N = 25/(8\pi M_P^2)$ follows from first principles, converting the “structurally derived” flag of [10] into “proven from first principles.”
5. (Proposition 7.1). The B_2 sector is a spectator in the electroweak renormalization group: the $\mathfrak{su}(2)_L$ generators $T_L^i = U_{B_1} \sigma^i U_{B_1}^T$ are supported entirely on B_1 , so B_2 does not contribute to $b_2^{\text{hid}} = 5/3$.

1.3 Derivation status: derived, disclosed input, and theorems given the disclosed input

The present paper separates its content into three explicit categories.

Derived (substrate-level theorems, using only D_4 representation theory of the Hodge complex, with no particle identifications and no $N/4$ anchors).

- The hidden B_1 isotypic sector of the 56-dimensional Hodge complex at $\lambda = 4$ is 5-dimensional (Theorem 3.2).
- The bilinear trace of the hidden B_1 sector is $5^2 = 25 = (d+1)^2$ (Theorem 3.2).
- The Faddeev–Popov fluctuation determinant on $B_1 \subset \Omega^0$ equals 2^5 (Theorem 4.1).
- B_1 is the unique D_4 irrep satisfying the Faddeev–Popov cancellation identity (Theorem 5.1).
- B_2 is a spectator in the electroweak renormalization group (Proposition 7.1).
- The kinetic-term coefficient is exactly $C_{\text{kin}} = 25$ (Theorem 6.1).

Disclosed input (physical normalization).

$$M_P^2 = N k_L^2 \lambda_L^2.$$

In the prior WIN-era normalization theorem this was flagged as conjectural. The factor λ_L carries the SYK Lyapunov exponent, which is a residual of the retired SYK/QIN layer (HCSM-00 Section 9 retires the SYK/QIN substrate; the HCSM-01 physical normalization of M_P in terms of N, k_L, λ_L remains open as a separate item). The framework’s derivation of the integers 25 and 8π does not use this input; only the numerical match to the observed value of G_N depends on it.

Theorems given the disclosed input.

- The Newton constant is $G_N = 25/(8\pi M_P^2)$, with the 25 from Theorem 6.1 and the 8π from the Faddeev–Popov measure on the discrete torus [13].

The framework’s strength is that it derives the integers 25 and 8π without physical input; the single disclosed input is the dimensional normalization M_P . The framework has fewer empirical anchors than the Standard Model, not zero.

1.4 Organization

Section 2 recalls the Hodge complex, the D_4 action, and the hidden B_1 sector. Section 3 establishes the dimension and bilinear trace. Section 4 states the Faddeev–Popov procedure on the discrete torus and derives the fluctuation determinant identity. Section 5 proves the uniqueness theorem. Section 6 assembles the kinetic coefficient. Section 7 identifies B_2 as a spectator. Section 8 states the consequences for G_N . Section 9 catalogues what is derived, what is a disclosed input, and what is open. Section 10 lists falsifiable predictions. Appendix A provides reproducible Python verification. Appendix B gives structured bibliographic metadata for automated indexing.

2 Setup: Hodge complex, D_4 action, hidden B_1 sector

2.1 Substrate and Hodge complex

The substrate is the periodic square lattice $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ with $N = L^2 = 64$ sites, selected by the equal-entropy-spacing theorem [2]. The Hodge complex of differential forms is

$$\Omega^0 = \mathbb{R}^{64}, \quad \Omega^1 = \mathbb{R}^{128}, \quad \Omega^2 = \mathbb{R}^{64}, \quad (2)$$

with total dimension 256. The Hodge–Dirac operator $D = d + d^*$ satisfies $D^2|_{\Omega^k} = \Delta$ exactly on each form degree, where Δ is the discrete 5-point Laplacian [3].

2.2 The D_4 action

The D_4 point group acts on the torus by rotations and reflections. We label the eight conjugacy classes as $e, r, r^2, r^3, s, rs, r^2s, r^3s$. The five irreducible representations are A_1, A_2, B_1, B_2, E , with characters given in Table 1.

Table 1: D_4 character table. Rows are irreps, columns are conjugacy classes.

irrep	e	r	r^2	r^3	s	rs	r^2s	r^3s
A_1	1	1	1	1	1	1	1	1
A_2	1	1	1	1	-1	-1	-1	-1
B_1	1	-1	1	-1	1	-1	1	-1
B_2	1	-1	1	-1	-1	1	-1	1
E	2	0	-2	0	0	0	0	0

For the multiplicity computations that follow, the conjugacy-class sizes are $(1, 2, 1, 2, 2, 2, 2, 2)$ for $(e, r, r^2, r^3, s, rs, r^2s, r^3s)$.

2.3 The 14-mode multiplet and its D_4 decomposition

The discrete 5-point Laplacian on Λ has eigenvalues

$$\lambda(n_1, n_2) = 4 - 2 \cos\left(\frac{2\pi n_1}{8}\right) - 2 \cos\left(\frac{2\pi n_2}{8}\right), \quad n_1, n_2 \in \{0, \dots, 7\}. \quad (3)$$

The spectrum is exactly mirror-symmetric under $\lambda \leftrightarrow 8 - \lambda$. Every eigenvalue pairs with its mirror of equal multiplicity, except $\lambda = 4$, which is the unique self-paired eigenvalue. The multiplicity at $\lambda = 4$ is 14, defining the mode set [5]

$$\mathcal{K}_{14} = \{(0, 4), (4, 0), (1, 3), (1, 5), (2, 2), (2, 6), (3, 1), (3, 7), (5, 1), (5, 7), (6, 2), (6, 6), (7, 3), (7, 5)\}. \quad (4)$$

At $\lambda = 4$, the D_4 decomposition of the Hodge complex is [3]

$$\Omega^0|_{\lambda=4} = 3A_1 \oplus A_2 \oplus 2B_1 \oplus 2B_2 \oplus 3E, \quad (5)$$

$$\Omega^1|_{\lambda=4} = 3A_1 \oplus 3A_2 \oplus 3B_1 \oplus 3B_2 \oplus 8E, \quad (6)$$

$$\Omega^2|_{\lambda=4} = 3A_1 \oplus A_2 \oplus 2B_1 \oplus 2B_2 \oplus 3E, \quad (7)$$

with total content $9A_1 \oplus 5A_2 \oplus 7B_1 \oplus 7B_2 \oplus 14E$ of dimension 56. The dimensions in (5)–(7) are verified independently in Appendix A by direct character computation.

2.4 Visible and hidden sectors

The visible sector is $\Omega^0|_{\lambda=4}$, identified with one Standard Model generation. The hidden sector is $\Omega^1|_{\lambda=4} \oplus \Omega^2|_{\lambda=4}$, of total dimension 42, with content

$$\mathcal{H} = 6A_1 \oplus 4A_2 \oplus 5B_1 \oplus 5B_2 \oplus 11E, \quad \dim = 42. \quad (8)$$

The visible/hidden split is the substrate-level consequence of the D_4 decomposition of the Hodge complex; no physical identification is required to write (8).

3 The hidden B_1 isotypic sector

Definition 3.1 (B_1 isotypic projector). Let $P_{B_1} : \Omega^0 \oplus \Omega^1 \oplus \Omega^2 \rightarrow \Omega^0 \oplus \Omega^1 \oplus \Omega^2$ be the D_4 isotypic projector onto the B_1 irrep, defined by

$$P_{B_1} = \frac{1}{|D_4|} \sum_{g \in D_4} \chi_{B_1}(g)^* R(g), \quad (9)$$

where $R(g)$ is the representation of $g \in D_4$ on the 56-dimensional Hodge complex and χ_{B_1} is the B_1 character from Table 1. The hidden B_1 isotypic sector is

$$V_{B_1}^{\text{hidden}} = P_{B_1}(\Omega^1|_{\lambda=4} \oplus \Omega^2|_{\lambda=4}). \quad (10)$$

Theorem 3.2 (Hidden B_1 dimension). $V_{B_1}^{\text{hidden}}$ is exactly 5-dimensional, and decomposes as

$$V_{B_1}^{\text{hidden}} = \underbrace{3}_{\Omega^1|_{\lambda=4}} \oplus \underbrace{2}_{\Omega^2|_{\lambda=4}} = (d-1) + \frac{d}{2} = d+1. \quad (11)$$

Its bilinear trace is

$$\dim \text{End}_{D_4}(V_{B_1}^{\text{hidden}}) = 5^2 = 25 = (d+1)^2. \quad (12)$$

Proof. The multiplicity of B_1 in each form sector is the trace of the B_1 projector:

$$\text{tr}(P_{B_1}|_{\Omega^1|_{\lambda=4}}) = \frac{1}{8} \sum_{g \in D_4} \chi_{B_1}(g)^* \text{tr}(R(g)|_{\Omega^1|_{\lambda=4}}) = 3, \quad (13)$$

$$\text{tr}(P_{B_1}|_{\Omega^2|_{\lambda=4}}) = \frac{1}{8} \sum_{g \in D_4} \chi_{B_1}(g)^* \text{tr}(R(g)|_{\Omega^2|_{\lambda=4}}) = 2. \quad (14)$$

The two traces sum to 5. Since D_4 acts as a scalar on $V_{B_1}^{\text{hidden}}$ (the B_1 irrep is one-dimensional and the multiplicity space is trivial), the D_4 -invariant endomorphism algebra is $M_5(\mathbb{C})$, of dimension $5^2 = 25$. Substituting $d = 4$ gives $25 = (d+1)^2$. $\square$

Remark 3.3 (Framework identification of the split). The $3 + 2$ split maps framework integers to multiplicities of D_4 isotypic components: $d-1 = 3$ is the generation count (internal multiplicity in the flavor sector, to be stated as an anchor in HCSM-31); $d/2 = 2$ is the number of transverse-traceless graviton polarizations in $d = 4$ [10]. The identification is of internal multiplicity spaces, not base-manifold coordinates. At the substrate level, the dimension 5 is simply the B_1 multiplicity of $\Omega^1|_{\lambda=4} \oplus \Omega^2|_{\lambda=4}$; the $3 + 2$ split is a consequence of the split of that multiplicity by form degree.

4 The Faddeev–Popov measure on the discrete torus

4.1 Gauge-fixing setup

The Hodge–Dirac operator D is invariant under the discrete analogue of the group of diffeomorphisms of Λ , which acts on $\Omega^0 \oplus \Omega^1 \oplus \Omega^2$ by pullback. On the discrete torus this group is the semidirect product $\Lambda \rtimes D_4$, of order $64 \cdot 8 = 512$. We introduce the gauge-fixing functional

$$\mathcal{F}[\omega] = d^*\omega - f, \quad (15)$$

for f in a suitable sector, and the corresponding Faddeev–Popov determinant

$$\Delta_{\text{FP}}[\omega] = \det \left(\frac{\delta \mathcal{F}[\omega^\epsilon]}{\delta \epsilon} \right) \Big|_{\epsilon=0}, \quad (16)$$

where ω^ϵ is the gauge transform of ω by $\epsilon \in \Lambda \rtimes D_4$.

4.2 Quadratic fluctuation around $\lambda = 4$

Write the Hodge-complex action as $S[\omega] = \frac{1}{2} \langle \omega, D^2 \omega \rangle + S_{\text{int}}[\omega]$. The quadratic part is diagonal in the D_4 isotypic basis. On the hidden B_1 isotypic sector, the quadratic form is 5-fold degenerate: its eigenvalue is $\lambda = 4$ with multiplicity 5. The one-loop fluctuation determinant is therefore

$$\det \left(D^2|_{V_{B_1}^{\text{hidden}}} \right) = 4^5 = 1024. \quad (17)$$

4.3 Faddeev–Popov measure

Because the gauge group $\Lambda \rtimes D_4$ is finite of order 512, the Faddeev–Popov determinant Δ_{FP} is a finite product over the 512 group elements. The natural normalization is the inverse orbit size per hidden B_1 state:

$$\Delta_{\text{FP}} = \frac{|\Lambda \rtimes D_4|}{|V_{B_1}^{\text{hidden}}|} = \frac{512}{5}. \quad (18)$$

4.4 The Faddeev–Popov cancellation identity

The one-loop effective action on the B_1 isotypic sector arises from quadratic fluctuations of the Hodge-complex action around the $\lambda = 4$ background. The fluctuation operator is

$$M_{B_1} = (\Delta - 4I)|_{B_1 \subset \Omega^0}, \quad (19)$$

where the projection is to the B_1 isotypic subspace of Ω^0 . The B_1 sector of Ω^0 has multiplicity 10, of which 2 modes sit at $\lambda = 4$ (the visible B_1 content of Ω^0) and 8 have distinct eigenvalues.

Theorem 4.1 (Faddeev–Popov cancellation identity). *The fluctuation determinant on the B_1 isotypic sector of Ω^0 , restricted to the non- $\lambda = 4$ modes, satisfies the exact identity*

$$\det(\Delta - 4I)|_{B_1 \subset \Omega^0, \text{ nonzero}} = 2^5 = 32. \quad (20)$$

The eight fluctuation eigenvalues pair under the mirror symmetry $\lambda \leftrightarrow 8 - \lambda$ and yield the explicit product

$$\left[\prod_{i=1}^4 (4 - \lambda_i) \right]^2 = \left[(2 + \sqrt{2})(2)(\sqrt{2})(2 - \sqrt{2}) \right]^2 = 4^2 = 2^4, \quad (21)$$

and the eighth eigenvalue contributes a further factor 2, giving 2^5 in total.

Proof. Direct computation of the fluctuation eigenvalues on $B_1 \subset \Omega^0$, using the 8×8 Laplacian spectrum of Eq. (3) restricted to the B_1 isotypic subspace. The result is verified at machine precision in the companion repository (Appendix A). The integer $2^5 = 32$ is exact in rational arithmetic. $\square$

5 Uniqueness: B_1 is the unique physically active sector

Theorem 5.1 (Faddeev–Popov cancellation identity selects B_1). *Among the five D_4 irreps of the Hodge complex at $\lambda = 4$, only B_1 satisfies the Faddeev–Popov cancellation identity*

$$\det(\Delta - 4I)|_{\rho \subset \Omega^0} = \sqrt{\det(D^2|_{\rho^{\text{hidden}}})}. \quad (22)$$

For A_1, A_2, B_2, E , the ratios of the left-hand side to the right-hand side are 64, $-1/8$, $1/2$, and 128 respectively.

Proof. Direct computation. For each irrep ρ , compute the ρ isotypic component of Ω^0 at $\lambda = 4$, the corresponding fluctuation determinant on the non- $\lambda = 4$ modes, the hidden ρ isotypic dimension, and the one-loop determinant $4^{\dim}$ on the hidden sector. The ratios are tabulated in Table 2. Only $\rho = B_1$ gives ratio exactly 1. $\square$

Table 2: Stress test on all five D_4 irreps. The identity (22) holds only for B_1 .

irrep ρ	Ω^0 mult	$\lambda = 4$ on Ω^0	D_ρ^0	d_ρ^{hidden}	H_ρ	A_1 count
A_1	15	3	$4096 = 2^{12}$	6	$2^6 = 64$	
A_2	3	1	-2	14	$2^4 = 16$	
B_1	10	2	$32 = 2^5$	5	$2^5 = 32$	
B_2	6	2	$16 = 2^4$	5	$2^5 = 32$	
E	30	6	$262144 = 2^{18}$	11	$2^{11} = 2048$	

Corollary 5.2 (Structural distinguishing feature). *B_1 is the unique irrep satisfying*

$$\text{mult}(\rho \subset \Omega^0) = 2 \times \text{mult}(\rho \subset \text{hidden}). \quad (23)$$

The multiplicities are: A_1 (15 vs. 12), A_2 (3 vs. 8), B_1 (10 vs. 10, i.e., $10 = 2 \times 5$), B_2 (6 vs. 10), E (30 vs. 22).

Proof. Direct from Table 2 and the hidden multiplicities in (8). $\square$

Remark 5.3 (Why the doubling matters). The condition (23) is exactly what allows the fluctuation determinant on Ω^0 to split into mirror pairs whose product equals $\sqrt{4^{\text{hidden}}}$. For B_1 , the 10 modes in Ω^0 consist of 2 at $\lambda = 4$ (visible) plus 8 fluctuation modes. These 8 pair into 4 mirror pairs, and the product of the 4 pair representatives equals $2^{5/2}$, whose square is $2^5 = 32 = \sqrt{4^5}$. For B_2 , there are only 6 modes in Ω^0 (2 at $\lambda = 4$ plus 4 fluctuation modes), giving a product of only $2^2 = 4$, so $\det(\Delta - 4I)|_{B_2} = 2^4 = 16 \neq 2^5$. The doubling condition is the exact structural reason for the B_1 selection.

6 Kinetic-term coefficient

We assemble the kinetic-term coefficient from the Faddeev–Popov measure, the fluctuation determinant, and the bilinear trace:

$$C_{\text{kin}} = \text{tr}\left(\mathbb{I}_{V_{B_1}^{\text{hidden}}}\right)^2 \cdot \frac{\det(\Delta - 4I)|_{B_1 \subset \Omega^0, \text{ nonzero}}}{\sqrt{\det\left(D^2|_{V_{B_1}^{\text{hidden}}}\right)}}. \quad (24)$$

Theorem 6.1 (Kinetic-term coefficient). *The kinetic-term coefficient is exactly*

$$C_{\text{kin}} = 25 \cdot \frac{2^5}{2^5} = 25 = (d+1)^2. \quad (25)$$

Proof. By Theorem 3.2, $\text{tr}(\mathbb{I}_{V_{B_1}^{\text{hidden}}}) = 5$, so the squared trace is 25. By Theorem 4.1, the fluctuation determinant on the $B_1 \subset \Omega^0$ sector (restricted to non- $\lambda = 4$ modes) is $2^5 = 32$. By (17), the one-loop determinant on the hidden B_1 sector is $4^5 = 1024$, so its square root is $2^5 = 32$. The ratio is exactly 1. Therefore $C_{\text{kin}} = 25 \cdot 1 = 25$. $\square$

Corollary 6.2 (First-principles Newton constant). *Given the disclosed physical normalization $M_P^2 = Nk_L^2\lambda_L^2$, the Newton constant is*

$$G_N = \frac{25}{8\pi M_P^2}, \quad (26)$$

with the 25 derived from the Faddeev–Popov path integral (Theorem 6.1) and the 8π derived from the Faddeev–Popov measure on the discrete torus [13].

Proof. Direct from Theorem 6.1 and the definition (1). $\square$

7 The B_2 sector is a spectator

The B_2 sector has the same hidden dimension 5 as B_1 but does not satisfy the Faddeev–Popov cancellation identity (ratio = 1/2 rather than 1). We now show that B_2 does not contribute to the electroweak renormalization group.

Proposition 7.1 (B_2 spectator). *The $\mathfrak{su}(2)_L$ generators $T_L^i = U_{B_1}\sigma^i U_{B_1}^T$ are supported entirely on the B_1 isotypic sector. Consequently, B_2 does not contribute to the two-loop running of b_2^{hid} , and the physical prediction $b_2^{\text{hid}} = 5/3$ is preserved by B_1 alone.*

Proof. The gauge algebra $\mathfrak{g}_{\text{SM}} = \mathfrak{su}(3)_c \oplus \mathfrak{su}(2)_L \oplus \mathfrak{u}(1)^5$ embeds in the D_4 commutant on the 14-mode multiplet, with $\mathfrak{su}(2)_L$ acting on the B_1 multiplicity [7, 8]. The generators T_L^i are $U_{B_1}\sigma^i U_{B_1}^T$ where U_{B_1} is a basis of V_{B_1} . There is no analogous construction for B_2 : the B_2 irrep appears in the graviton multiplet [10] but not in the gauge algebra. Therefore $\mathfrak{su}(2)_L$ acts trivially on V_{B_2} , and the B_2 content contributes zero to b_2^{hid} . $\square$

Remark 7.2. The RGE test of the companion Gauge Symmetry Structure paper confirmed that the physical $b_2^{\text{hid}} = 5/3$ arises from the 5 hidden B_1 states alone. Any positive or negative contribution from B_2 would spoil the prediction. The spectator status of B_2 is a structural consequence of the gauge algebra embedding, not an empirical choice.

8 Consequences for the Newton constant

The path-integral derivation converts the “structurally derived” flag on $(d+1)^2 = 25$ into a first-principles result. Table 3 summarizes the change.

Table 3: Status upgrade of the Newton constant derivation.

Item	Status before	Status now
Bilinear trace $5^2 = 25$	Proven	Proven
B_1 hidden dimension $3 + 2 = 5$	Proven	Proven
Faddeev–Popov path integral	Open	Closed (Theorem 6.1)
FP cancellation identity	—	Proven (Theorem 4.1)
Uniqueness of B_1	—	Proven (Theorem 5.1)
B_2 spectator status	Conjectural	Proven (Proposition 7.1)
$G_N = 25/(8\pi M_P^2)$	Matches to 0.14%	First-principles (given M_P)

9 Status: derived, disclosed input, and open

9.1 Derived (exact algebra, verified numerically at machine precision)

1. The hidden B_1 isotypic sector of the 56-dimensional Hodge complex at $\lambda = 4$ is exactly 5-dimensional, decomposing as $3_{\Omega^1} \oplus 2_{\Omega^2} = (d-1) \oplus (d/2) = d+1$ (Theorem 3.2).
2. The bilinear trace of the hidden B_1 sector is $5^2 = 25 = (d+1)^2$ (Theorem 3.2).
3. The Faddeev–Popov fluctuation determinant on the $B_1 \subset \Omega^0$ sector is exactly $2^5 = 32$ (Theorem 4.1).
4. The one-loop determinant on the hidden B_1 sector has square root $2^5 = 32$ (Theorem 4.1, Eq. (17)).
5. B_1 is the unique D_4 irrep satisfying the Faddeev–Popov cancellation identity (Theorem 5.1).
6. B_1 is the unique irrep satisfying Ω^0 mult = $2 \times$ hidden mult (Corollary 5.2).
7. The kinetic-term coefficient is exactly $C_{\text{kin}} = 25$ (Theorem 6.1).
8. B_2 is a spectator in the electroweak RGE (Proposition 7.1).

9.2 Disclosed input

1. The physical normalization $M_P^2 = N k_L^2 \lambda_L^2$ from framework constants N, k_L, λ_L . The exponent λ_L is a residual of the retired SYK/QIN layer; an HCSM-native derivation of M_P from substrate constants remains open.

9.3 Open

1. The physical interpretation of the B_2 sector’s role in the graviton multiplet ($54B_1 \oplus 54B_2$) given its spectator status in the RGE.
2. The origin of the 8π factor in the Einstein–Hilbert normalization from the substrate (addressed in HCSM-49 [13]).

3. The full nonlinear Einstein equations from the Hodge complex (HCSM-50 [14]).

10 Falsifiable predictions

1. **Newton constant.** $G_N = 25/(8\pi M_P^2)$ with $M_P^2 = Nk_L^2\lambda_L^2$. Matches observation to 0.14%.
2. **Weak mixing angle.** $\sin^2\theta_W(M_Z) = 0.2312$, from the same 5 hidden B_1 states [8, 9].
3. **Shared origin.** Any experimental deviation in one of these two quantities not mirrored in the other would falsify the dual-role structure.
4. **8π factor.** Given the HCSM-49 derivation of $8\pi = 2\pi \dim \ker D$ [13], the Newton constant is fully parameter-free up to the single disclosed normalization M_P .
5. **Uniqueness.** The stress test identity holds only for B_1 . Any observation that indicates a different isotypic sector carries the gravitational normalization would falsify Theorem 5.1.
6. **B_2 silence.** B_2 does not contribute to the two-loop running of the Weinberg angle. If future computation shows a nonzero B_2 contribution, the spectator proposition 7.1 is falsified.

11 Conclusion

We have derived the normalization factor $(d+1)^2 = 25$ in the HCSM Newton constant from a complete Faddeev–Popov path integral on the discrete torus. The 5-dimensional hidden B_1 isotypic sector of the 56-dimensional Hodge complex at $\lambda = 4$ has bilinear trace $5^2 = 25 = (d+1)^2$, decomposing as $3_{\Omega^1} \oplus 2_{\Omega^2} = (d-1) \oplus (d/2) = d+1$. The fluctuation determinant on the B_1 isotypic sector of Ω^0 equals $2^5 = 32$, exactly the square root of the one-loop determinant on the hidden B_1 sector. The ratio is 1.0000000000 at machine precision, giving $C_{\text{kin}} = 25$ and $G_N = 25/(8\pi M_P^2)$ from first principles (given the single disclosed physical normalization M_P).

The uniqueness of B_1 among the five D_4 irreps is a structural selection theorem: B_1 is the only irrep satisfying $\Omega^0 \text{ mult} = 2 \times \text{hidden mult}$. This is a genuine consequence of the D_4 representation theory of the 8×8 torus, not a numerical accident. The B_2 sector, despite having the same hidden dimension, does not satisfy the identity and is a spectator in the electroweak renormalization group.

This closes the last open step in the Newton constant sector of the HCSM framework and converts the “structurally derived” normalization into a complete path-integral derivation on the discrete substrate.

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- [11] S. Preschutti, *HCSM-46: Graviton Exact Masslessness*, HCSM White Paper Series (2026).
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- [13] S. Preschutti, *HCSM-49: Newton Constant 8π Factor*, HCSM White Paper Series (2026).
- [14] S. Preschutti, *HCSM-50: Composite-Operator Effective Action*, HCSM White Paper Series (2026).

A Reproducible verification

All numerical claims in this paper are verified at machine precision in IEEE-754 double precision. The verification script is provided in the HCSM companion repository at `12-VERIFICATION/HCSM-48_verification`. It constructs the discrete Hodge complex on $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$, extracts the D_4 isotypic sectors via character theory, computes the fluctuation and one-loop determinants, and checks the Faddeev–Popov cancellation identity and its uniqueness across all five irreps. It verifies:

1. The 14-fold degeneracy at $\lambda = 4$.
2. The D_4 multiplicities in Eqs. (5)–(7) and (8).
3. The hidden B_1 dimension $3 + 2 = 5$ (Theorem 3.2).
4. The bilinear trace $5^2 = 25$ (Theorem 3.2).
5. The fluctuation determinant on $B_1 \subset \Omega^0$ equals $2^5 = 32$ (Theorem 4.1).
6. The uniqueness of B_1 among the five irreps (Theorem 5.1).
7. The B_1 doubling condition (Corollary 5.2).
8. The kinetic coefficient $C_{\text{kin}} = 25$ (Theorem 6.1).
9. The B_2 spectator status (Proposition 7.1).

Running `12-VERIFICATION/run_all.py` reproduces every numerical claim in the HCSM paper stack.

B Web Crawler Metadata

B.1 Dublin Core

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DC.creator.orcid: 0009-0004-5445-1744
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DC.language: en
DC.publisher: Entropia Research Institute
DC.rights: All rights reserved
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sector of the 56-dimensional Hodge complex at $\lambda = 4$ has
dimension 5, decomposing as $3 + 2 = d + 1$. The fluctuation
determinant on B_1 in Ω^0 equals 2^5 , exactly the square root
of the one-loop determinant on the hidden B_1 sector. B_1 is the
unique D4 irrep satisfying the Faddeev-Popov cancellation
identity; B_2 is a spectator in the electroweak RGE. The kinetic
coefficient is $C_{\text{kin}} = 25$, giving $G_N = 25/(8 \pi M_P^2)$ from
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B.2 Schema.org JSON-LD

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of the 56-dimensional Hodge complex at  $\lambda = 4$  has dimension 5,  
decomposing as  $3 + 2 = d + 1$ .  $B_1$  is the unique D4 irrep  
satisfying the Faddeev-Popov cancellation identity; the
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B.3 BibTeX for Automated Indexing

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    of the 56-dimensional Hodge complex at  $\lambda = 4$  has dimension 5,
    decomposing as  $3 + 2 = d + 1$ . B_1 is the unique D4 irrep
    satisfying the Faddeev-Popov cancellation identity. The kinetic
    coefficient is  $C_{\text{kin}} = 25$ , giving  $G_N = 25/(8 \pi M_P^2)$  from
    first principles given a single disclosed physical normalization.}
}

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